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A neural network application to subcontractor rating in construction firms

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In construction companies, subcontractor rating, due to its intrinsic ambiguity and difficult formalization, is a particularly complex task, usually accomplished by management experience and intuition. In this paper, a neural network is proposed to support management in subcontractor rating. Neural networks, being able to learn directly by examples the managers' logic, are suitable to support the solution of this type of problem. By an application case related to an assembly operation in a construction site, the neural network implementation and the related managerial and technical innovations are investigated. © 1997 Elsevier Science Ltd and IPMA

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Facing a competitive and dynamic environment, large construction firms are forced into more complex and appropriate organization structures.¹

The large size and the complexity of construction projects, located in different sites and made of several production activities which involve different specializations and technical capabilities, usually drive construction firms to operate as general contractors resorting to specialized outside firms to carry out specific project activities. An efficient and effective coordination among all the project activities can be achieved by the definition of long term cooperations of general contractors with specialized outside firms.^{2,3} In fact, giving in subcontract often a large part of their works (usually more than 50% of their income), construction companies have their performance strongly dependent on subcontractors. General contractors need then to establish a reliable network of subcontractors, so that the design and management of the subcontractor network becomes a strategic issue.

In a competitive bidding for the subcontracting of a part of a project, the general contractor management has to rate the competitors considering several information related to the specific project, to the managerial and technical skills of the potential subcontractors and, of course, to the contents of the bids.

Uncertainty and ambiguity are deeply rooted in the specific transaction and the decision-making process involves complex problems of analysis and judgment.⁴

Some quantitative models have been proposed to help general contractor management in dealing with the bid rating problem.⁵ These models, however, even when assisted by information systems more oriented to propose the solution of management problems, such as Decision Support Systems (DSS) or Expert Systems (ES), generally do not succeed in transferring properly the experts' judgment, mainly due to the not transparent logic of the decision process.⁶

A more suitable tool to support subcontractor rating in construction firms can be based on neural networks. They can be considered systems able to cope with decision-making problems not structurable and highly affected by uncertainty and ambiguity.^{7,8} A peculiar property of neural networks is their possibility of reproducing the expert's behavior by learning through examples. These systems, then, introduce technological as well as organizational innovations in management decision-making.

In this paper, an application of neural networks to support general contractor management in subcontractor rating is discussed, pointing out the peculiarities and potentialities of the approach and its impact on the organization of the decision process.

Subcontractor rating in construction firms

Subcontractor rating is usually applied by general contractor management for the selection of a specialized firm to carry out part of a project. The rating is generally based on the evaluation of both the bids received

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by the potential subcontractors and the competitors' characteristics, such as reliability, size, etc.

In the subcontractor rating process it is not always easy to recognize precise rules, but there is, in general, a coherent way to solve the problem. This coherence can be rooted in intuition, experience, common sense, or not explicit rules. Subcontractor rating is then a problem usually solved by subjective criteria, based on personal experience and belief, on the available information and, sometimes, on techniques and algorithms (such as the weighted sum) supporting the decision process.

From a methodological point of view, the process of subcontractor rating can be schematized in three main phases:

- (a) choice of the rating parameters to be assessed;
- (b) parameter evaluation;
- (c) competitor evaluation.

In the phase (a), according to the specific project objectives and contingencies, the decision-maker selects the main parameters to be assessed for the subcontractor choice. These parameters can be split in elementary parameters according to a hierarchical approach,⁵ even if managers usually prefer to deal with few parameters, mainly for two reasons: (a) they like to evaluate a limited number of qualitative parameters, and (b) they need to remain more confident with the rating process, in particular if it is based on computer systems.

The phase (b) is characterized by the decision-maker evaluation of the parameters. In particular, judgments can be expressed either by votes (for instance, quantitatively, by numbers between 1 and 10 or, qualitatively, by adjectives such as low, medium, high), or by comparative sentences, such as «A is *lightly better* than B». If the parameters are organized by a hierarchical structure, the evaluation of the main parameters, depending on the evaluation of the elementary parameters, is obtained by the decision-maker expertise and/or by the support of specific techniques and algorithms, such as weighted sums, fuzzy logic, expert systems, etc.

In the phase (c), the competitor rating is determined by the analysis of the parameters previously assessed. As in the evaluation of the dependent parameters in a hierarchical structure (phase b), this analysis can be supported by the same type of techniques and algorithms.

The use of algorithms to support subcontractor rating is generally aimed to achieve a formalization of the decision-making process that can allow a more objective and reliable evaluation, particularly complex when several parameters have to be considered. These algorithms, however, are often affected by the subjectivity of the evaluation procedures. For example, procedures based on the use of weighted sums or even more complex approaches, such as the analytical hierarchical process (AHP)⁵ or the fuzzy logic,⁹ try to identify with rigid algorithms a behavioral rule (the expert's judgment) which is often not explicit in the reality.

To support subcontractor rating, then, a new approach, based on neural networks (NN), can be adopted. In fact, neural networks can learn to rate subcontractors directly by examples of rating, without resorting to the formalization of the decision-maker expertise in rules. In this way, the hard formalization

of mental processes, which are very seldom explicitable, is not necessary.

Neural networks

In this paper, a *back propagation* neural network¹⁰ is proposed to support subcontractor rating. This type of *supervised* neural network allows the transfer of knowledge simply by the definition of examples, which provide the specific decision-making process with a very user-friendly approach.

Focusing on the management point of view, the main phases of the NN implementation—after the problem analysis, involving the definition of the variables and environment of the specific problem—can be schematized in the NN design, programming and testing.

The design of the neural network can be considered a 'basic programming' and can be split in two activities: the definition of the network architecture and the definition of the network internal rules.

The NN architecture is made of nodes and connections between nodes. Every node belongs to a node line (layer), while connections usually link the nodes of two adjacent layers. The node number of each layer generally depends on the specific application. The nodes of the input layer, for instance, can represent the different parameters of the problem: the parameter values are the input data for the network. The nodes of the output layer can be characterized, for instance, by a single neuron providing a numeric evaluation. For the hidden layers (usually one or two), an effective number of nodes is usually determined by trials, since there is no certain rule to determine it.¹⁰

The internal rules define the neural network behavior. Besides the activation law, which regulates the signal transmission among nodes, a fundamental rule is the learning law which, during the programming phase, models the interactions among nodes in order to provide the proper answer. The back propagation is a typical learning law for networks trained by examples made by inputs and outputs known a priori (supervised learning).¹¹

The NN programming (or 'learning phase') consists in defining a *training set* of examples suitable to make the network learn the decision-maker's behavior. By the training set, the network learns the way to provide a response to the information (input) introduced by the users. The training set definition is then particularly crucial for the correct working of the network. The variety of the examples included in the training set has generally to be wide. In fact, by a wide learning the network can correctly process other different cases, later investigated by the users. With a low number of examples, instead, it does not succeed in generalizing properly. Moreover, eventual expert's incoherence, i.e. the definition of different outputs in correspondence to similar inputs, negatively affects the reliability of the network response.

Finally, the NN testing consists in the network performance checking. Among the main NN performance, generalization, i.e. the capacity of providing an appropriate response in correspondence of new information, and noise tolerance, i.e. the ability of recognizing unclear information, are critically analyzed in this phase.

A first test can be done analyzing the network responses to the inputs of the training set. In this case, the obtained outputs are compared with the outputs of the training set. Another test can be made analyzing the network performance in correspondence of some examples (*validation set*) not used in the learning phase.

Testing the accuracy of the NN approach by a comparison with other approaches, such as the AHP and the fuzzy logic, seems not simply accomplishable. In fact, differently from the NN, the simulation of the expert's reasoning by other approaches requires the definition of specific algorithms that, aimed to formalize the expert's logic, are arbitrary in nature. Each approach could then allow several comparisons with the NN, so that the evaluation of the NN performance would depend on the specific algorithm and approach selected for the comparison.

The three phases of the neural network implementation are closely interrelated. In particular, for example, the testing phase can put in evidence problems associated to the inadequacy of either the learning phase or the design phase.

Finally, the neural network implementation usually requires other auxiliary phases, such as the software choice, the NN integration with other information systems, the NN maintenance, which are not discussed in this paper.

An application case

To show the practical implementation of a neural network to subcontractor rating, an application case is analyzed, with the support of the purchasing director of a general contractor mostly operating in large civil works projects. The NN implementation requires the definition of the problem environment, the parameters involved in a decision process, their measure, and the desired type of rating.

The problem environment is determined by the kind of activity to be given in subcontract, as well as by the project contingencies characterizing the scenario in which the activity has to be carried out. In this application case, the expert has been invited to identify a typical construction activity usually given in subcontract and two of the most common situations in which it can occur. The activity considered is an assembly operation in a construction site, characterized by not relevant technical contents. In a first scenario (situation A), the activity belongs to the critical path of the project. In a second scenario (situation B), the activity is not critical and the project is characterized by a low expected profit for the general contractor. The two different scenarios influence the neural network implementation, because the neural network has to be trained in different ways, depending on the specific situation, in order to reproduce different decision-maker's behaviors.

To evaluate the specialized firms applying as potential subcontractors (competitors), the purchasing director's suggestions have lead to consider the following parameters:

- p_1 : price reduction
- p_2 : time reduction
- p_3 : technical/qualitative characteristics of the bid

- p_4 : contractual reliability of the competitor
- p_5 : management skills of the competitor.

These five parameters, according to the expert's opinion, meaningfully and summarily express the characteristics of each bid and of the related competitor. Considering the specific assembly characteristics in the two situations, the most important parameters for the expert's evaluation are, in the first situation, time reduction (and, to a lower extent, contractual reliability and price), while, in the second situation, price reduction.

The type of measure for price and time reductions is substantially different from the one that characterizes the other three parameters. In fact, reductions are measured by the percentages of the initial price (and time) defined by the client for the bid. On the contrary, technical/qualitative level of the bid, contractual reliability and management skills of the specialized firms are evaluated qualitatively. In particular, the expert has proposed four possible verbal values for these parameters: excellent (A); good (B); sufficient (C) and insufficient (D). For the NN implementation, however, since the network can only process numerical input values, a numerical scale (such as $A = 4$, $B = 3$, $C = 2$, $D = 1$) has to be associated to the verbal values considered.

The set of competitors that participate in the bid can be classified according to either an ordinal scale or an interval scale. Since a measure of the 'distance' among the evaluations of the different competitors is considered useful to the decision-maker and the back propagation NN is suitable to provide an interval scale classification, this classification is preferred in the application case. Once the values of the parameters concerning each competitor (inputs) are defined, the neural network can then provide a numeric evaluation x of the competitor (output), ranging from 0 to 10.

The network architecture is designed (Figure 1) according to the type of data (five parameters) and response (a numerical value) required by the application. An input layer made of five nodes (a node for each parameter) and an output layer made of one node, providing the competitor evaluation, are then considered. One hidden layer made of seven nodes has also been selected after several trials during the testing

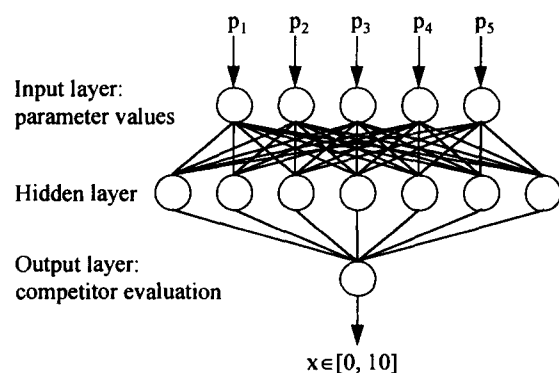


Figure 1 Neural network architecture for the application case

Table 1 Training set for the supervised learning of the neural network

Competitors Parameters	1	2	3	4	5	6	7	8	9	10
p ₁ : price reduction (%)	30	5	10	8	7	5	20	17	7	25
p ₂ : time reduction (%)	30	5	10	27	8	18	10	5	12	5
p ₃ : technology and quality	A	A	B	B	A	B	C	B	C	D
p ₄ : contractual reliability	A	B	A	D	B	C	D	A	B	B
p ₅ : management skills	A	A	B	A	C	B	C	C	D	C
Expert—Situation A	10	4	5	8	4	6	4	4	4	3
Expert—Situation B	10	4	5	4	4	3	5	6	3	7

Table 2 Testing of the neural network on the validation set

Competitors Parameters	I	II	III	IV	V
p ₁ : price reduction (%)	10	20	22	14	6
p ₂ : time reduction (%)	15	10	5	7	20
p ₃ : technology and quality	B	C	C	D	A
p ₄ : contractual reliability	C	D	B	A	D
p ₅ : management skills	D	B	B	C	C
Expert—Situation A	5	4	3	3	6
Neural network—Situation A	4.6	5	4.1	3.3	6.6
Expert—Situation B	4	6	7	5	3
Neural network—Situation B	3.8	5.8	7	5	3

phase. The software adopted in this application is NeuralWorks Professional II, by NeuralWare Inc.

Since the neural network emulates the expert's behavior by learning how he usually solves the rating problem, the expert has defined 10 examples for each of the two different situations, A and B. His judgments (outputs), used for the supervised learning of the neural network, are shown in *Table 1*, where Competitor 1 represents an optimal reference competitor for the expert.

As it can be noticed in *Table 1*, the examples of the training set are differently evaluated by the expert in the two different situations. The network has then to be differently trained in the two situation A and B, in order to learn how to properly answer in each specific situation. A node assuming two different values in correspondence of the two situations A and B could be added to the input layer, in order to train the network once for both situations. However, for this application case, the network configuration has not been changed in order to separately treat the two situations, so that eventual network modifications generated by inefficacy in any of the two situations cannot affect the other one.

Once the learning phase is completed, the testing phase can be executed. A measure of the NN effectiveness can be provided by the mean square error MSE:

$$MSE = \frac{\sqrt{\sum_{i=1}^n (x_i - E(i))^2}}{n}$$

where n is the number of examples to be evaluated ($n = 10$ in the training set), x_i is the network output related to the example i ($i = 1, \dots, n$), and $E(i)$ is the correspondent expected value, i.e. the expert evaluation.

The first test consists in the analysis of the network responses to the input data of the training set. In this case, the network produces outputs that are identical with the expert's evaluations ($MSE = 0$). For this reason they are not reported in *Table 1*. The test is

then extended to a new set of examples (validation set), shown in *Table 2*. In this table, the expert's appraisals and the network responses are compared to evaluate the network effectiveness.

In *Table 2* it can be observed that in the situation B the network has a high capability of generalizing, since it succeeds in reproducing not only the right ranking of competitors of the validation set, but also, with a good approximation ($MSE = 0.06$), the numeric appraisals of the expert.

In the situation A, instead, the network results are not very close to the expert's judgments ($MSE = 0.34$). In particular, the network provides a wrong response even about the ranking of the validation set, for instance classifying Competitor II in a higher position than Competitor I, contrasting the expert's opinion. As a consequence, the network responses cannot be considered dependable.

No significant performance improvement in terms of MSE and ranking is obtained by the variation of the number of hidden nodes and layers (network configurations characterized by 5 up to 10 hidden nodes as well as by two hidden layers have been considered). Also different durations (between 5.000 and 50.000 epochs) of the training phase, tested to verify the efficacy of the network training, have not yielded better results.

It is then inferred that the difference between the situations A and B probably lies on a more complex recognition (by the network) of a basic coherence in the expert's reasoning for the situation A. In this situation, in fact, even attributing more importance to the

Table 3 Further examples of the training set for the situation A

Competitors Parameters	11	12	13	14	15
p ₁ : price reduction (%)	5	10	9	2	12
p ₂ : time reduction (%)	10	15	22	15	5
p ₃ : technology and quality	A	B	B	A	D
p ₄ : contractual reliability	C	B	D	B	A
p ₅ : management skills	B	A	D	D	A
Expert—Situation A	4	6	6	5	3

Table 4 Testing of the neural network (with the enlarged training set) on the validation set for the situation A

Competitors Appraisals	I	II	III	IV	V
Expert—Situation A	5	4	3	3	6
Neural network—Situation A	5	4	3.1	2.9	6.3

parameter 'time reduction', the expert is also sensitive to other parameters (such as price), but his appraisal rules are not clearly explicable. To improve the network behavior, it is then necessary to provide a wider training set, by which the network can better learn the expert's belief. To this purpose, the expert is required to enlarge the previous training set for the situation A by five further examples, reported in *Table 3*. In real cases, for instance when historical data are not available, these further examples can be based on the educated guess and beliefs of the expert.

In *Table 4* it can be observed how, using the enlarged training set for the situation A, the network response to the validation set are undoubtedly closer to the judgments formulated by the expert (MSE = 0.06). The network seems then to have more effectively learnt the expert's reasoning.

Once the testing phase has been successfully completed, the network is ready for the use. Given a real set of competitors, then, the decision-maker can be supported in the subcontractor rating by assessing the parameters previously considered and obtaining the final competitor rating by the network.

Management implications and conclusions

The complexity of subcontractor rating, due to the uncertainty and ambiguity involved in the decision-making process, requires a formalization aimed to reduce the expert's subjectivity. A neural networks application to subcontractor rating has been proposed to support general contractor management more effectively than other techniques, which appear not very suitable to this task.

The NN capability of seizing knowledge by examples and not by rules represents a very interesting and innovative factor even in terms of managerial innovation. In fact, at the management level, some logic and psychological barriers limiting the adoption of other information systems can be removed. Focusing on examples lets the project manager and his team spend more time on the meaning and the metrics of the subcontractor evaluation rather than on the definition of the appraisal algorithms. This work involves the commitment of a team, whose discussion results enhance a common understanding and a more complete assessment of the subcontractor performance.

An application case, related to the evaluation of potential subcontractors competing for a bid related to an assembly activity in a construction site, has pointed out the technical and managerial characteristics of the implementation. The absence of formal mechanisms to support the evaluation, enabling the expert to express his opinions without transforming or adjusting them according to the support tool requirements, represents an extremely innovative characteristic for the specific case analyzed.

By the analysis of the application case, the user-friendliness of the approach comes out during both the

learning phase, developed only by examples, and the network use, which requires just the expression of simple judgments about the competitors' characteristics.

It has been pointed out, however, that the learning process can be affected by the complexity of the decision-making environment and/or by the possible incoherence introduced by the expert's examples. In these cases, it is necessary to extend the learning phase by the adoption of further examples, improving the knowledge representation.

It has also been emphasized that the learning phase of a neural network is referred to a specific decision-making context: when it changes, it is necessary to differently train the network. In addition, even the NN architecture is referred to a specific scenario, which depends, for instance, on the peculiar characteristics of the activity to be given in subcontract and on the number of parameters to be evaluated for each competitor.

Hence, the use of other information systems, such as DSS or ES, can be suggested to support neural networks during the typical phase of computer implementation, for instance related to the definition of the network configuration for each specific situation, allowing a higher flexibility in the NN utilization for different problems and environments. In particular, this information system integration could be specifically advisable when neural networks are designed to support more complex subcontractor rating models, which can characterize some real projects. For instance, if hierarchical dependent parameters are considered, the NN approach potentiality, from the management point of view, is particularly interesting, since the user can be required to evaluate only elementary characteristics of the competitors, while the network can directly provide the final subcontractor rating.

However, recent results related to the implementation of a NN-based DSS in a construction company have stressed the increasing relevance of subcontractor evaluations based only on the bidding price and on qualitative and sometimes intangible assessments, such as contractual reliability and quality performance. The implementation has shown that experts require tools to integrate such evaluations, but they prefer to operate only with few variables, in order to remain confident with the tool and verify its consistency with their assessments.

All these aspects provide interesting issues for further investigation in this research area.

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